AIR FUEL RATIO (A/F) SENSOR [SKYACTIV-G 1.3, SKYACTIV-G 1.5]

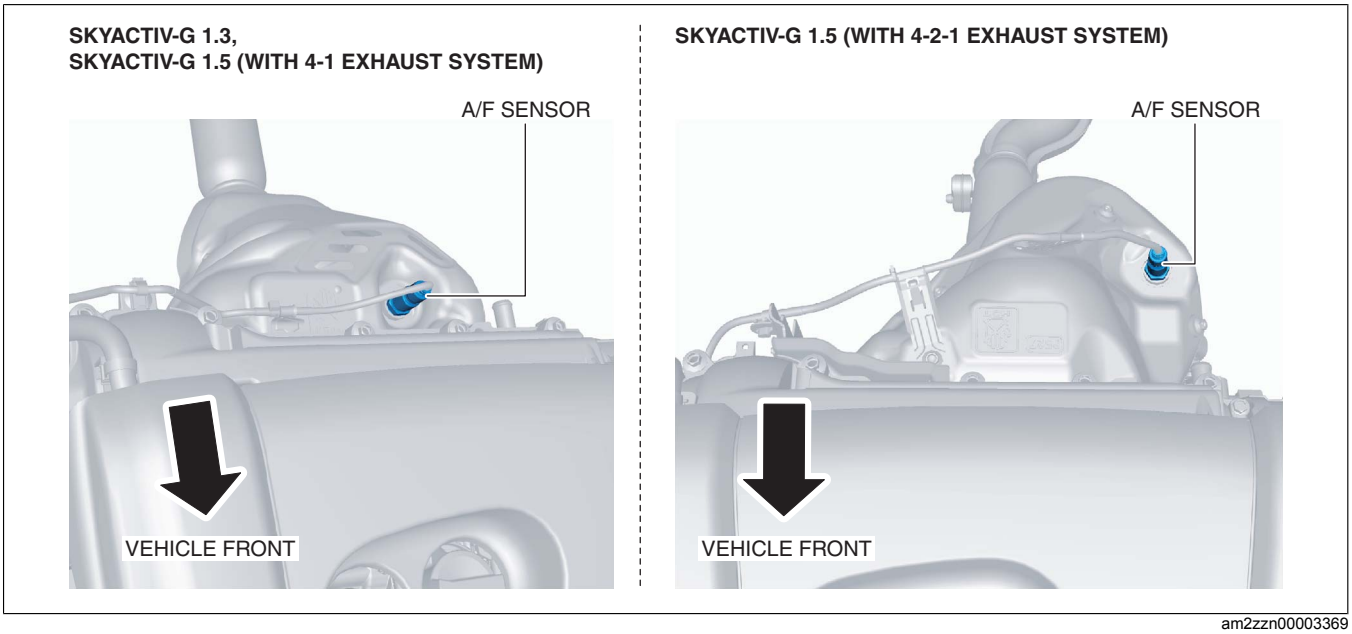
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Purpose/Function

- Detects the oxygen concentration (all-range air/fuel ratio) in the exhaust gas as basic information (mainly used to determine the fuel injection amount) and sends it to the PCM as an oxygen concentration (all-range air/fuel ratio) signal.
- Because the A/F sensor has an output characteristics proportional to the actual air/fuel ratio, a detailed air/fuel ratio can be detected and compared using the HO2S, which detects the rich/lean of the oxygen concentration in relation to the theoretical air/fuel ratio.

Construction

- Installed to the exhaust manifold.

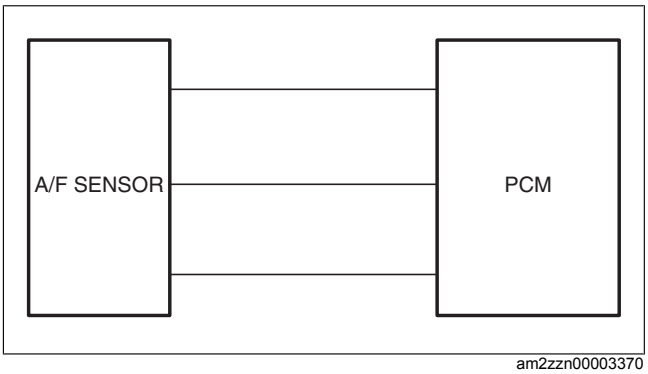


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- A zirconia element has been adopted (as the detecting element) for detecting the oxygen concentration (all-range air/fuel ratio).

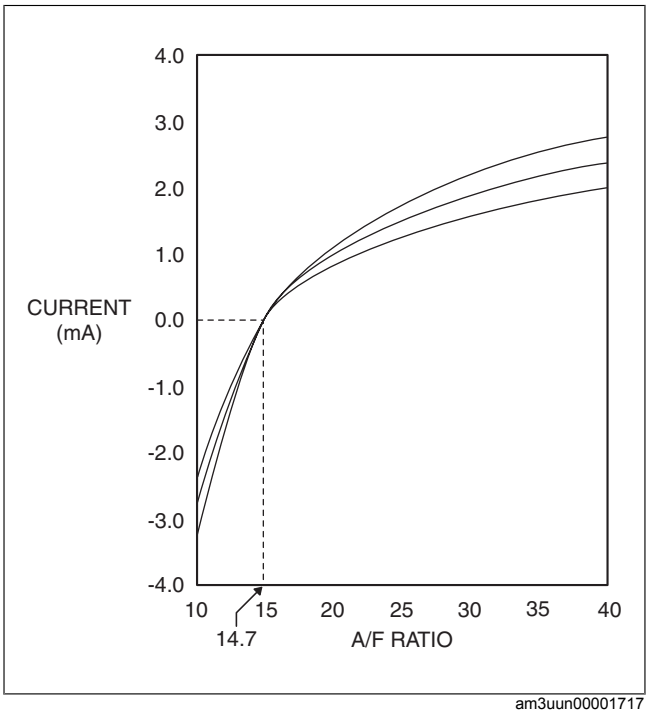
Operation

- The A/F sensor has an oxygen standard chamber, which is always kept at **100%** of oxygen concentration, and an oxygen concentration detection chamber, where the oxygen concentration in the exhaust gas is detected.
- A zirconia element has the characteristic of moving oxygen when electrical current flows. The oxygen standard chamber is kept at **100%** oxygen concentration using this characteristic, and by comparing it with the oxygen concentration inside the oxygen concentration detection chamber, the oxygen concentration in the exhaust gas can be detected.



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A/F sensor characteristic



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Fail-safe

DTC No.	Fail-safe function
P0031:00	• Stops the fuel feedback control.
P0032:00	• Stops the fuel feedback control.
P0130:00	• Fixes duty value of A/F sensor heater. • Stops fuel feedback control of A/F sensor.
P0131:00	• Fixes duty value of A/F sensor heater. • Stops fuel feedback control of A/F sensor.
P0132:00	• Fixes duty value of A/F sensor heater. • Stops fuel feedback control of A/F sensor.
P0133:00	• Fixes duty value of A/F sensor heater • Stops fuel feedback control of A/F sensor
P0134:00	• Fixes duty value of A/F sensor heater. • Stops fuel feedback control of A/F sensor.
P2237:00	• Fixes duty value of A/F sensor heater. • Stops fuel feedback control of A/F sensor.